

Member ID: _____

Time: _____

Rank: _____



COMPUTER NETWORK TECHNOLOGY

(300)

STATE 2023

TOTAL POINTS (50 x 2 points each)

_____ (100 points)

Test Time: 60 minutes

GENERAL GUIDELINES:

Failure to adhere to any of the following rules will result in disqualification:

1. Member must hand in this test booklet and all printouts if any. Failure to do so will result in disqualification.
2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests (handwritten, photocopied, or keyed) are allowed in the testing area.
3. Electronic devices will be monitored according to ACT standards.

Choose the *best* answer to the following questions. Mark answers clearly if using a scantron form. Mark A if the statement is true. Mark B if the statement is false.

1. Which of the following is a valid DHCP operation that occurs during a client negotiation?
 - A. Repeat
 - B. Assess
 - C. Obtain
 - D. Discover

2. Which of the following describes a technology that allows access to memory directly, without involving the processor, cache, or operating system?
 - A. NPIV
 - B. RDMA
 - C. NPAR
 - D. MIMO

3. What feature uses DHCP to provide a specific address to a network printer based on its MAC address?
 - A. IP Options
 - B. DHCP Relay
 - C. Reservation
 - D. Lease

4. What type of cable would be used to connect like devices without auto-sensing ports?
 - A. Straight-through
 - B. Crossover
 - C. Rollover
 - D. Patch

5. Which of the following best describes DHCP snooping?
 - A. Inspects and monitors DHCP traffic, but does not interfere
 - B. Captures and forwards all DHCP traffic to the DHCP sanitation server
 - C. Excludes rogue DHCP servers and removes malicious / malformed DHCP traffic
 - D. Penetrates security vulnerabilities in an existing DHCP infrastructure

6. Which of the following describes what the abbreviation “MU-MIMO” stands for?
 - A. Multi-Unit, Multiple-Ingress, Multiple-Obfuscation
 - B. Multi-User, Multiple-Input, Multiple-Output
 - C. Multi-Upload, Multiple-Inbound, Multiple-Outbound
 - D. Multi-Unload, Multiple-Instance, Multiple-Overwrite

7. Which of the following describes a Standard Access Control List?
 - A. A list of permit or deny rules by source IP address
 - B. A list of whitelisted domains
 - C. A list of blacklisted domains
 - D. A list of permit or deny rules by source and destination IP addresses
8. Which of the following is **not** a layer of the TCP/IP model?
 - A. Application
 - B. Transport
 - C. Data Link
 - D. Network Interface
9. Which of the following topologies offers the lowest level of redundancy?
 - A. Full mesh
 - B. Hub-and-spoke
 - C. Ring
 - D. Partial mesh
10. NFS is primarily used by which of the following clients?
 - A. Linux
 - B. Microsoft
 - C. macOS
 - D. Android
11. Which type of network is based on network clients sharing resources with one another?
 - A. Client-server
 - B. Client-client
 - C. Peer-to-peer
 - D. Peer-to-server
12. Which of the following is the **last** usable host address in the following network:
192.168.0.63/26?
 - A. 192.168.0.190
 - B. 192.168.0.254
 - C. 192.168.0.158
 - D. None of the above

13. Which of the following server types functions as a relay, commonly for web requests?
- A. Proxy server
 - B. RDP server
 - C. VPN server
 - D. Exchange server
14. What is the default mask for a Class B network?
- A. 255.0.0.0
 - B. 255.255.0.0
 - C. 255.255.255.0
 - D. 255.255.255.255
15. A user is attempting to access an external website via HTTPS. The browser is reporting that the certificate is invalid, however the website was not reporting any issues the day before. Which of the following is the most likely culprit?
- A. The root certificate is not trusted
 - B. The hostname does not match what is listed on the certificate
 - C. The certificate has expired
 - D. The certificate or a certificate in the chain is on the CRL
16. Which of the following is the subnet mask for CIDR notation /18?
- A. 255.254.0.0
 - B. 255.255.128.0
 - C. 255.255.192.0
 - D. 255.255.255.224
17. Which of the following technologies has helped alleviate the depletion of the IPv4 address space?
- A. DHCP
 - B. NAT
 - C. Wi-Fi
 - D. RIP
18. What is the technology invented by Cisco that uses the concept of a default gateway address assigned to a virtual router?
- A. CARP
 - B. DDR
 - C. VRRP
 - D. HSRP

19. Which of the following describes an architecture / technology that was introduced to make a network infrastructure more dynamic and manageable?
- A. WWW
 - B. CRC
 - C. RADIUS
 - D. SDN
20. Which of the following would most likely be the cause of a broadcast storm?
- A. Spanning tree protocol
 - B. Link aggregation
 - C. Switching loop
 - D. All of the above
21. Which of the following is a valid FQDN for a server named “FileServer” in the BPA.org domain?
- A. BPA.FileServer
 - B. FileServer.BPA.org
 - C. FileServer@BPA.org
 - D. BPA.org.FileServer
22. What protocol is used to request a MAC address that corresponds to a known IPv4 address on the local network?
- A. IGMP
 - B. ARP
 - C. TTL
 - D. ICMP
23. What is the purpose of having a quorum, with regards to a failover cluster?
- A. Provides a shared folder for transferring files between nodes
 - B. Facilitates live migration between cluster nodes
 - C. Helps maintain data consistency across nodes
 - D. None of the above
24. Which of the following is a common type of social engineering attack?
- A. Twinning
 - B. Phishing
 - C. DDoS
 - D. Ghosting

25. Which of the following is a valid type of virtual switch within Hyper-V?
- A. Egress network
 - B. Public network
 - C. Private network
 - D. Distributed network
26. A network administrator changed the default listening port of Remote Desktop Protocol to 3390 on the Remote Desktop server (hostname: RDServer, IP address: 192.168.1.10). When specifying the Remote Desktop server and port number through RDP, which of the following is not considered valid?
- A. RDServer:3390
 - B. 192.168.1.10:3390
 - C. RDServer -P 3390
 - D. all are valid
27. A specific client cannot reach their internal website. All other clients can reach the internal website without issue. Given the following potential issues, which is the most likely culprit?
- A. The client default gateway is incorrect
 - B. The server default gateway is incorrect
 - C. The client firewall is blocking port 443
 - D. The server firewall is blocking port 443
28. What is the primary file-sharing protocol in Windows?
- A. SSJ
 - B. SMB
 - C. S2D
 - D. SQL
29. A bridge operates at which layer of the OSI model?
- A. Layer 1
 - B. Layer 2
 - C. Layer 3
 - D. Layer 4
30. Which is a valid APIPA address?
- A. 169.255.215.12
 - B. 169.254.255.58
 - C. 169.250.250.25
 - D. 169.254.248.58

31. A server would likely use ____ to connect to a NAS array:
- A. iSCSI
 - B. SAS
 - C. SMB
 - D. SFTP
32. A server would likely use ____ to connect to a SAN array:
- A. iSCSI
 - B. SAS
 - C. SMB
 - D. SFTP
33. A server would likely use ____ to connect to a DAS array:
- A. iSCSI
 - B. SAS
 - C. SMB
 - D. SFTP
34. How many bits dictate the Vendor ID in a MAC address?
- A. 18
 - B. 28
 - C. 24
 - D. 16
35. 20. What is the CIDR notation for the following subnet mask: 255.255.255.192?
- A. /24
 - B. /26
 - C. /28
 - D. /20
36. What is one of the primary advantages of using a small form-factor pluggable (SFP)?
- A. Prevents ports from causing a loop
 - B. Provides flexibility
 - C. Allows bridging between multiple ports
 - D. Prevents jumbo frames / packets from exceeding ~9000 bytes
37. Which of the following ranges is valid for private usage?
- A. 10.0.0.0 - 10.255.255.255
 - B. 172.16.0.0 - 172.32.0.0
 - C. 192.164.0.0 - 192.164.255.255
 - D. All of the above

38. Which of the following are valid fiber optic connectors?
- A. RJ-11 and RJ-45
 - B. FC and SC
 - C. VGA and DVI-D
 - D. DE-9 and DB-25
39. What is the term for the weakening of a signal over distance?
- A. Fade
 - B. Crosstalk
 - C. Attenuation
 - D. Short
40. What is the broadcast address for the following host: 192.168.0.167/24?
- A. 192.168.0.255
 - B. 192.168.0.175
 - C. 192.168.0.160
 - D. 255.255.255.0
41. Which of the following statements is **true** when considering the primary differences between a proxy server and VPN?
- A. VPNs encrypt traffic
 - B. Proxy servers encrypt traffic
 - C. VPNs do not mask IP addresses
 - D. All of the above
42. What transport layer protocol features connectionless communications?
- A. UDP
 - B. ICMP
 - C. IGMP
 - D. All of the above
43. Which of the following will prevent a DHCP server from assigning a specific IP address or address range within a DHCP scope?
- A. DHCP lease
 - B. DHCP registration
 - C. DHCP exemption
 - D. DHCP exclusion

44. As a network administrator, you are troubleshooting within an environment that uses DHCP. The issue is that some clients are receiving IP addresses on the 192.168.1.0/24 network, while some are receiving IP addresses on the 10.10.10.0/24 network. Which of the following is the most likely culprit?
- A. DNS is misconfigured
 - B. Rogue DHCP server
 - C. Incorrect default gateway
 - D. Subnet mask is misconfigured
45. Users would like to access server resources from home (shared folders, application databases, etc.). Which of the following would be a secure option?
- A. On the router, forward port 3389 to the server to allow external RDP connectivity
 - B. Connect the server directly to the ISP device, bypassing the router, so that it receives an external IP address
 - C. Allow users to share a login to third party, remote connectivity software
 - D. Configure Remote Desktop Gateway
46. What protocol is commonly used to carry voice over IP traffic through the network?
- A. RTM
 - B. IMAP
 - C. RTP
 - D. ICMP
47. Which of the following network topologies is least suited to small business and home networks?
- A. Star
 - B. Bus
 - C. Hybrid
 - D. Mesh
48. You are troubleshooting an issue in which all clients lost internet connectivity after adding a new client to the network. The IP address assigned to the new client is 10.10.10.1/24. The clients regain internet connectivity if the new client is shut down, what is the most likely culprit?
- A. IP address conflict with the domain controller
 - B. IP address conflict with the default gateway
 - C. DNS is misconfigured
 - D. DHCP is misconfigured

49. Which of the following network functions can help with network congestion by limiting the data transmission rate of the sender?
- A. Auto negotiation
 - B. Error control
 - C. Flow control
 - D. Auto sensing
50. What port number is used by HTTPS?
- A. 443
 - B. 125
 - C. 80
 - D. 23